

Weekly Reading Update on Transformer Foundations and Multi-Agent LLM Routing

Group 2 – CSE499A, Section 15

Faculty: MSRB

Mahamudul Islam (2312314042), Zawed Bin Tariq (2232811642),

Sayed Mortuza (2311448042), Md Farhan (2312703042)

Abstract—This weekly update summarizes Group 2’s first reading progress. Mahamudul Islam and Zawed Bin Tariq studied *MasRouter: Learning to Route LLMs for Multi-Agent System*, while Sayeed Mortuza and Md Farhan studied *Attention Is All You Need*. The objective was to understand both the foundation of Transformer-based LLMs and the recent idea of cost-aware routing in multi-agent LLM systems.

I. READING ASSIGNMENT

TABLE I
MEMBER-WISE WEEKLY READING TASK

Member	NSU ID	Assigned Paper
Mahamudul Islam	2312314042	MasRouter
Zawed Bin Tariq	2232811642	MasRouter
Sayed Mortuza	2311448042	Attention Is All You Need
Md Farhan	2312703042	Attention Is All You Need

II. PAPER 1: MASROUTER

MasRouter studies routing for LLM-based Multi-Agent Systems (MAS). Existing LLM routing usually selects one suitable LLM for a query, but a MAS requires more decisions. It must determine the collaboration mode, the number of agents, the role of each agent, and the LLM backbone assigned to each role. The authors define this broader problem as Multi-Agent System Routing (MASR) [1]. In simple terms, instead of always using the largest model or a fixed group of agents, *MasRouter* builds a customized team for each query.

MasRouter works through a cascaded controller network. First, a collaboration determiner selects a communication pattern such as chain, tree, complete graph, reflection, or debate. Second, a role allocator assigns suitable roles such as programmer, math analyst, physicist, critic, or test analyst. Third, an LLM router selects the most appropriate LLM for each agent according to the task and role. The main contribution is balancing performance and cost. The paper reports that *MasRouter* performs strongly across MMLU, GSM8K, MATH, HumanEval, and MBPP, while reducing unnecessary token and API overhead compared with several single-agent, multi-agent, and router baselines [1]. This is important because practical AI systems need both accuracy and efficiency.

III. PAPER 2: ATTENTION IS ALL YOU NEED

Attention Is All You Need introduced the Transformer architecture, which later became the basis of modern LLMs. Before this paper, many sequence models used RNNs, LSTMs, GRUs, or CNNs. These models were effective but often slow for long sequences because they processed tokens sequentially

or through limited local windows. The Transformer removes recurrence and convolution and uses self-attention to connect all tokens directly [2].

The central operation is scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \quad (1)$$

Here, Q , K , and V mean query, key, and value matrices. QK^T measures similarity between tokens, softmax converts those scores into weights, and the weighted values form the output representation. The scaling term $\sqrt{d_k}$ keeps the scores stable when the vector dimension becomes large. The model also uses multi-head attention so that different heads can learn different relationships in parallel. Positional encoding is added because the model has no recurrent order-tracking mechanism.

IV. COMBINED UNDERSTANDING

The two papers are connected. The Transformer paper explains how a single LLM-style model processes information internally using attention. *MasRouter* explains how multiple LLM-powered agents can be selected and coordinated externally for a task. Therefore, our reading covered both model architecture and system architecture. From the Transformer paper, we learned attention, multi-head attention, positional encoding, encoder-decoder design, and parallel training. From *MasRouter*, we learned MASR, collaboration topology, role allocation, model routing, and cost-performance trade-off.

V. WEEKLY OUTCOME AND NEXT PLAN

This week helped us build the theoretical base for understanding modern LLM and agentic AI systems. For next week, we will prepare a glossary of key terms, draw simplified architecture diagrams for Transformer and *MasRouter*, compare single-agent and multi-agent pipelines, and discuss how routing or attention-based models may support our CSE499A project direction.

REFERENCES

- [1] Y. Yue *et al.*, “MasRouter: Learning to Route LLMs for Multi-Agent System,” in *Proc. 63rd Annual Meeting of the Association for Computational Linguistics*, 2025.
- [2] A. Vaswani *et al.*, “Attention Is All You Need,” in *Proc. Advances in Neural Information Processing Systems*, 2017.